BP Internet Banking

Solution Architecture Proposal

Online and mobile banking platform: movement history, own-account and interbank transfers and payments, mobile onboarding with facial recognition.

Solution Architect practical exercise · November 2026

C4 model (Structurizr DSL) · AWS · Ecuador regulatory context

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1. Executive summary

BP (assumed to be **Banco Pichincha**, the largest private bank in Ecuador, ~6.5 M customers, ~5.2 M mobile-app users) needs an internet banking platform for a web SPA and a multiplatform mobile app. Customers check their movement history and make own-account and interbank transfers and payments; new customers open accounts from the app with facial recognition. Every movement must be notified, every action audited, and the platform must be highly available, secure and low-latency. The bank's 2025 app outages (supervised by the Superintendencia de Bancos, followed by a deposit run) make **resilience the first design driver**.

Key decisions

Area	Decision
Cloud	**AWS**, primary **us-east-1** (lowest measured latency from Ecuador, ~90 ms), **warm standby in us-east-2**; hybrid link to the on-premises Core with 2 Direct Connect links + VPN
Architecture style	**Microservices by bounded context** (14 domain services, coarse-grained), BFF per channel, event-driven support services
Front-end	**Angular** SPA (no tokens in the browser, BFF token handler) and **Flutter** mobile app (compared with React Native)
Authentication	OAuth 2.0 **Authorization Code + PKCE + PAR**, **FAPI 2.0** profile, **DPoP** sender-constrained tokens; biometric login through a **device-bound key** in the Secure Enclave / StrongBox; **step-up with dynamic linking** for transfers to new beneficiaries
Onboarding	Specialist **eKYC vendor** (FacePhi preferred, Jumio alternative, PoC) + **Registro Civil** biometric validation + AML screening, orchestrated as a **Saga**; no face templates stored
Integration	**Kong** API Gateway (single entry point), **Amazon MSK (Kafka)** event backbone, **EKS** compute, anti-corruption layers to the Core, complementary system and BCE payment network (**ISO 20022** `pacs.008` / `pacs.002`)
Frequent clients	**CQRS read model + cache-aside + event-driven invalidation + refresh-ahead** (dashboard p95 < 500 ms, Core protected)
Audit	Outbox + Kafka → Audit Service → **S3 Object Lock (WORM)** + hot index, **hash chain + signed digests**
Notifications	**SMS + email simultaneously** (regulation), 2 SMS providers with automatic failover, push as complement
Resilience	Multi-AZ, warm standby DR (**RTO 15 min / RPO 1 min** for critical journeys), auto-healing, chaos and DR drills
Cost	AWS ≈ **USD 105–120 k / month** (≈ USD 0.02 per active user); **SMS is the dominant running cost**

All decisions are recorded as **21 ADRs** (section 17), each with the alternatives evaluated and at least two justifications. The C4 model is a single **Structurizr DSL** workspace in the repository (architecture/workspace.dsl), so every diagram is generated from one consistent model.

2. Context, requirements and assumptions

2.1 Functional requirements

ID	Requirement	Source
FR-01	Mobile onboarding with **facial recognition**	Statement §3.5
FR-02	Login with username and password through the bank's **OAuth 2.0** product (SPA and mobile)	§3.4
FR-03	**Biometric login** after onboarding (fingerprint / Face ID)	§3.5
FR-04	**Movement history**	§3
FR-05	**Basic customer data** (Core) and **detailed data** (complementary system)	§3.1
FR-06 / 07 / 08	**Own-account and interbank transfers and payments**	§3
FR-09	**Notify every movement** through **at least 2** notification systems	§3.2
FR-10	**Audit database** with every customer action	§3.6
FR-11	**Persistence for frequent clients**, based on design patterns	§3.6
FR-12	**Integration layer with an API Gateway** routing by transaction type	§3.7
FR-13 – 15	*(added)* Transfer status tracking, beneficiary management, step-up confirmation	Implied
Regulatory additions	Device binding (transactions only from authorized devices), real-time risk scoring	Section 3

2.2 Quality attribute scenarios (selection)

ID	Attribute	Scenario → response measure
QAS-01	HA	An AZ fails → no visible outage; **99.99%** monthly for login, balance, transfers; 99.95% for the rest
QAS-02	DR	Primary region lost → critical journeys **RTO ≤ 15 min, RPO ≤ 1 min**
QAS-03	Fault tolerance	Core slow/down → circuit opens, balances and movements served "as of hh:mm", transfers rejected cleanly
QAS-04	Fault tolerance	Primary notification provider fails → delivered by the secondary in **≤ 60 s**, none lost
QAS-05	Fault tolerance	Customer retries a transfer → **0 duplicate debits** (idempotency key)
QAS-06 / 07 / 08	Performance	Frequent-client dashboard p95 **< 500 ms**; 3-month history p95 **< 1.5 s**; transfer accepted p95 **< 3 s**
QAS-09	Scalability	Payday at **3× peak** (~500 TPS monetary, ~5,000 req/s at the gateway) with error rate < 0.1%
QAS-10	Security	Stolen access token can't move money
QAS-11	Security	Presentation attack during onboarding is rejected (ISO/IEC 30107-3)
QAS-12	Auditability	Insider tries to change audit records → impossible (WORM) and detected
QAS-13	Observability	Error spike → alert in **≤ 5 min**, trace shows the failing dependency
QAS-14	Auto-healing	Crashed or hung instance replaced in **≤ 2 min** without human action
QAS-15	Modifiability	New channel or service added without changing existing services

ID	Attribute	Scenario → response measure
QAS-16	Cost	Night traffic ~10% of peak → compute scales down

2.3 Assumptions (grounded in public data on Banco Pichincha)

ID	Assumption
A-00 / 01	BP = Banco Pichincha; jurisdiction **Ecuador** (Superintendencia de Bancos, LOPDP)
A-02 / 03	~~6.5 M registered and **5.5 M active** digital users, **~~2 M daily active**
A-04 / 05	~~16 k concurrent sessions (80 k on payday); **~~160 TPS** normal, **500 TPS** design target for monetary transactions; **~5,000 req/s** at the gateway (public data: 164 M mobile transactions a month)
A-06	~90 / 10 read/write ratio of API calls
A-07 / 08	Availability and RTO/RPO as in QAS-01 / 02
A-09	Transaction history and audit **12 months online**, **10 years archive** for KYC/AML records (to confirm in the UAFE regulation)
A-10	"Payments" = to third-party and interbank accounts and the customer's own cards and loans; bill payments are a future extension
A-11	Core and complementary system stay **on-premises**, expose REST/SOAP APIs, can't be modified, and have limited capacity
A-12	The bank's OAuth 2.0 product supports OIDC, PKCE, MFA, custom claims and JWKS
A-13	Interbank: BCE **SPI** (deferred) + **real-time payments** (instant, capped at one basic salary)
A-14	~**90% mobile / 10% web** traffic
A-15	The bank's DEUNA wallet is out of scope, but the API layer must be reusable by it

Out of scope: card issuing and PCI DSS (no cards involved), loan origination, the identity product's internals, the Core's internals.

3. Regulatory considerations (Ecuador)

Ref	Norm	Authority	What it drives
R1	**Operational risk management norm** (SB, chapter replaced in 2023, reformed by SB-2024-02855)	Superintendencia de Bancos	Electronic-channel security, cloud and outsourcing, continuity, incident reporting
R3	**Ley Orgánica de Protección de Datos Personales (LOPDP)**	SPDP	Personal data, processors, breaches
R4 / R5	SPDP-SPD-2026-0039-R (**biometric data**) and SPDP-SPD-2026-0040-R (**breach notification**)	SPDP	Facial recognition, incident deadlines
R7	**AML law** (2024) and its 2026 regulation	UAFE	KYC at onboarding, transaction monitoring, record retention
R8	**BCE SPI technical norm** and **real-time payments messaging standard** (Res. BCE-GG-007-2026)	Banco Central del Ecuador	Interbank payments in ISO 20022
—	International references: ISO 27001/27017/27018, ISO 22301, COBIT 2019, OWASP ASVS/MASVS, FAPI 2.0, NIST SP 800-207	—	Security, continuity, governance

Requirements that shape the design

Requirement	Design response	Section
Notify every monetary transaction by **mobile messaging and email simultaneously**; notify every login	Notification service, SMS + email in parallel	11
2-of-3-factor authentication at login and for transfers to unregistered accounts	OIDC + MFA; device key + biometrics; step-up with dynamic linking	6
Mobile transactions **only from authorized devices**	Device binding with DPoP keys in secure hardware	6
Block after **3 failed attempts**, idle timeout, show last login	IdP lockout + notification, BFF sessions, profile	6
Real-time risk profiling of transactions	Risk Scoring service before money moves	12
Anti-malware / anti-tampering on channels	RASP, attestation, WAF, bot control	5, 12
12-month transaction history with IP, OTP used, notified contacts; **staff access logs**	Audit solution	10
Encryption, masking of account numbers, masked data outside production	KMS, field-level encryption, masking, separate accounts	9, 12
Cloud: inform the SB, **Tier III / TIA-942**, **ISO 27001/27017/27018**, annual audit, contract clauses, alternate providers	AWS attestations (AWS Artifact), DPA, two providers for critical services	3, 12
Continuity: RTO/RPO per critical process, annual tests, remote site not sharing the primary's risks	Warm standby in a different region, DR drills twice a year	13
Incidents: SB report **≤ 5 days**, root cause **≤ 10 days**; personal-data breach: processor → bank **≤ 2 days**, bank → SPDP **≤ 5 days**	SIEM, runbooks, provider contract clauses	12, 14
Biometrics: DPIA every 12 months, explicit consent, **non-biometric alternative**, templates over raw images, no solely automated decisions, no minors	Onboarding design, manual review queue, password + OTP always available	7

Requirement	Design response	Section
KYC before account opening, transaction monitoring, ≥ 10-year records	Onboarding Saga, AML Monitoring, WORM archive	7, 10
ISO 20022 `pacs.008` / `pacs.002`, debit before sending, irrevocability	Interbank Connector, transfer Saga	8

> Article numbers of the SB norm should be checked against the current consolidated text (the chapter was replaced in 2023); the 10-year AML retention should be confirmed in the UAFE regulation.

4. Architecture overview and C4 diagrams

> All diagrams are generated from one Structurizr model and embedded as **vector graphics**: zoom in for detail. Full-resolution PNG and SVG files are in the repository (`architecture/.structurizr/1/images/`).

4.1 Architecture style (ADR-004)

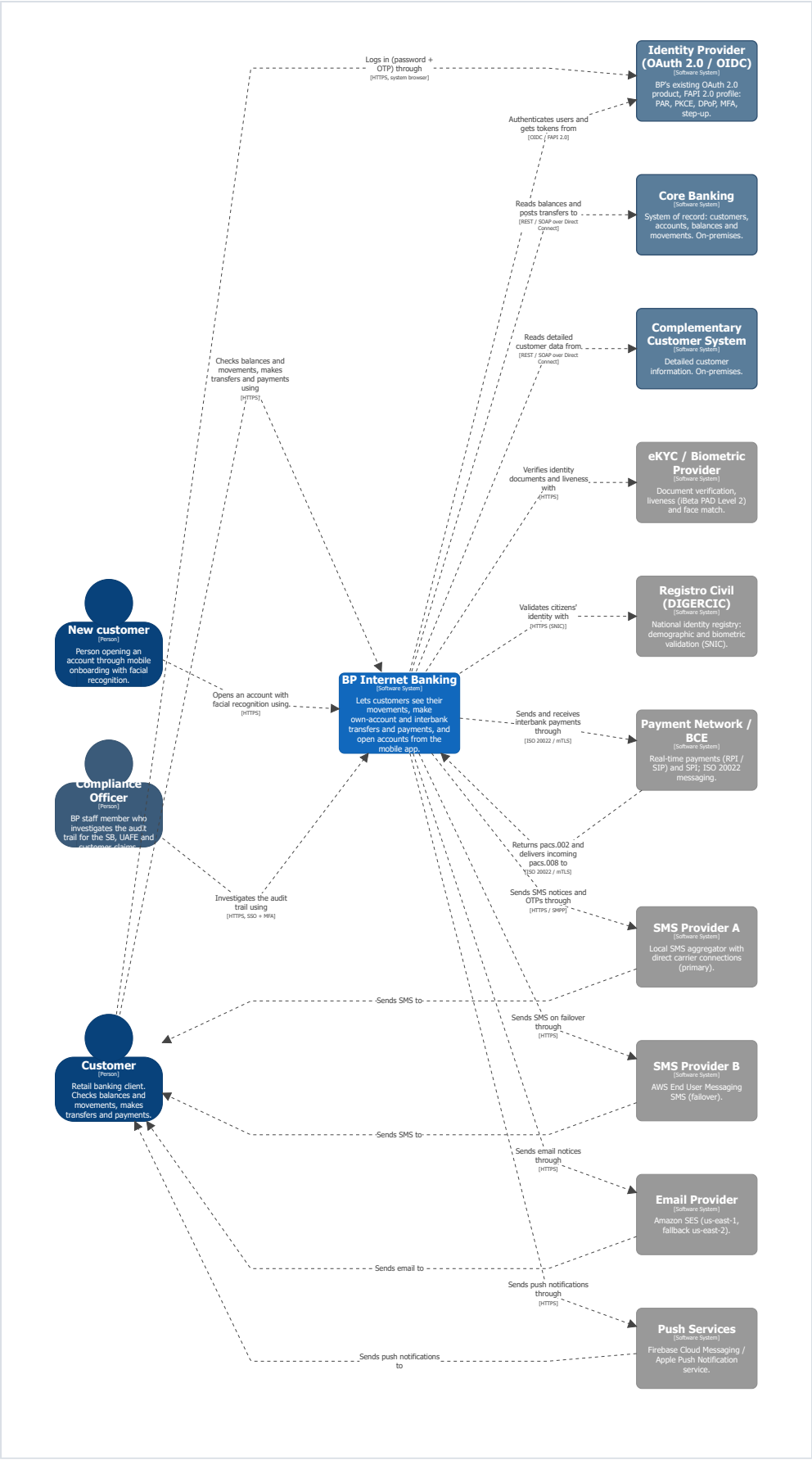
The platform is split into **10 bounded contexts** implemented as **14 coarse-grained microservices**, plus a channel layer (two BFFs) and shared platform services (gateway, event broker, cache, observability). A new service is created only when it has its own data, its own load profile or its own compliance rules. Communication is **synchronous** where the user waits or a decision is needed before money moves, and **asynchronous** (Kafka events) for notifications, audit, AML and read models, so those can never slow down or break a transfer.

Request path: **Edge (CloudFront + WAF) → API Gateway (Kong) → BFF (web / mobile) → domain services (mTLS mesh) → adapters → Core / external systems.**

Base service from the statement	Implemented by
Basic customer data query	**Customer Profile Service** (+ Customer Info Adapter for detailed data)
Movements query	**Movements Service** (CQRS read model)
Transfers (calling external services by type)	**Transfer Service** + **Interbank Connector**
Added for performance	Accounts Service (balances, dashboard cache), Movements read model, Core CDC Connector
Added for regulation	Device, Beneficiary & Limits, Risk Scoring, AML Monitoring, Onboarding
Added for decoupling	Notification, Audit, Core Adapter, Customer Info Adapter

4.2 Level 1: System context

For non-technical readers: who uses the platform and which BP and external systems it depends on.



Element	Role
Customer, New customer	Use the web and mobile apps; new customers onboard from the app
Compliance Officer	BP staff who investigate the audit trail

Element	Role
Core Banking, Complementary Customer System	BP on-premises systems of record (existing)
Identity Provider	BP's existing OAuth 2.0 / OIDC product
eKYC / Biometric Provider, Registro Civil	Identity verification during onboarding
Payment Network / BCE	Interbank real-time payments and SPI
SMS Providers A and B, Email Provider, Push Services	Notification channels

4.3 Level 2: Containers

The full container view is in the repository (02-Containers.svg). For readability it is presented here as five focused views of the same model.

Figure 4.3a – Channels, edge, API Gateway and BFFs

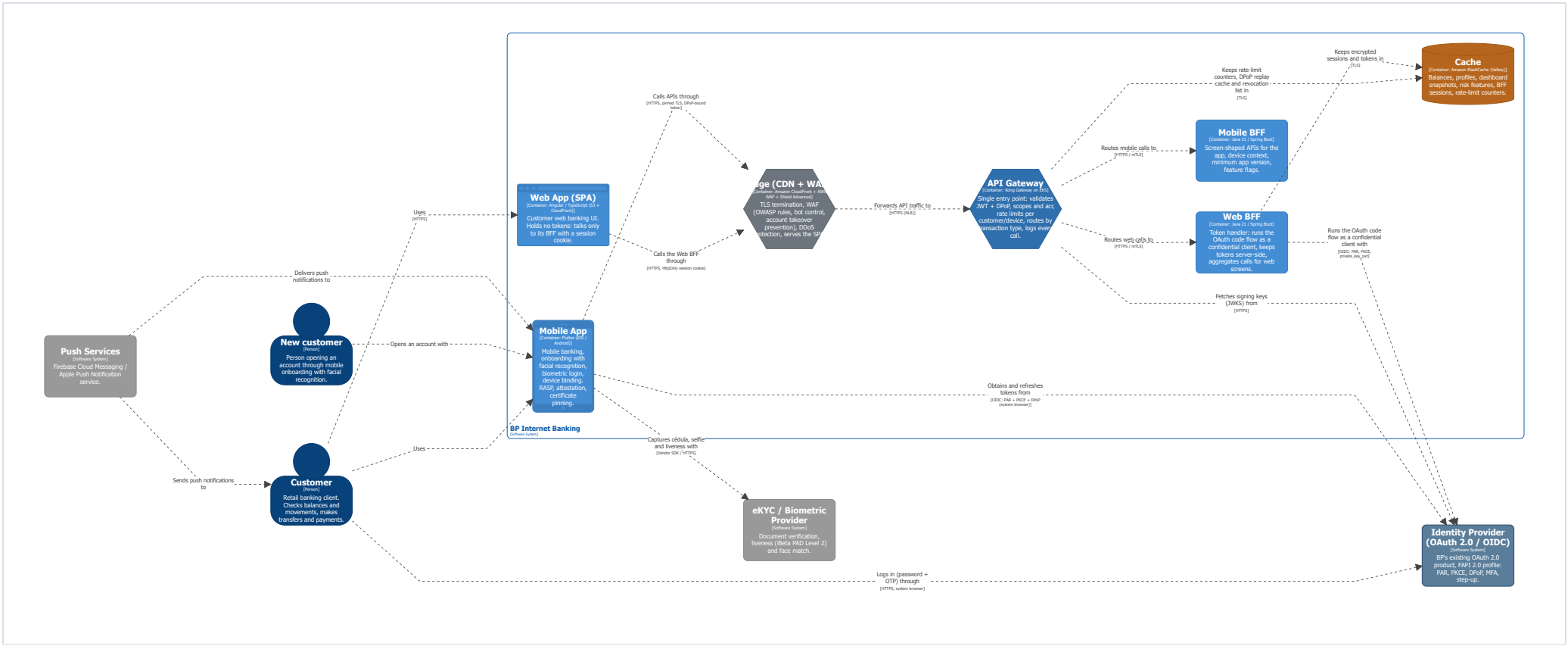


Figure 4.3b – Transfers and payments

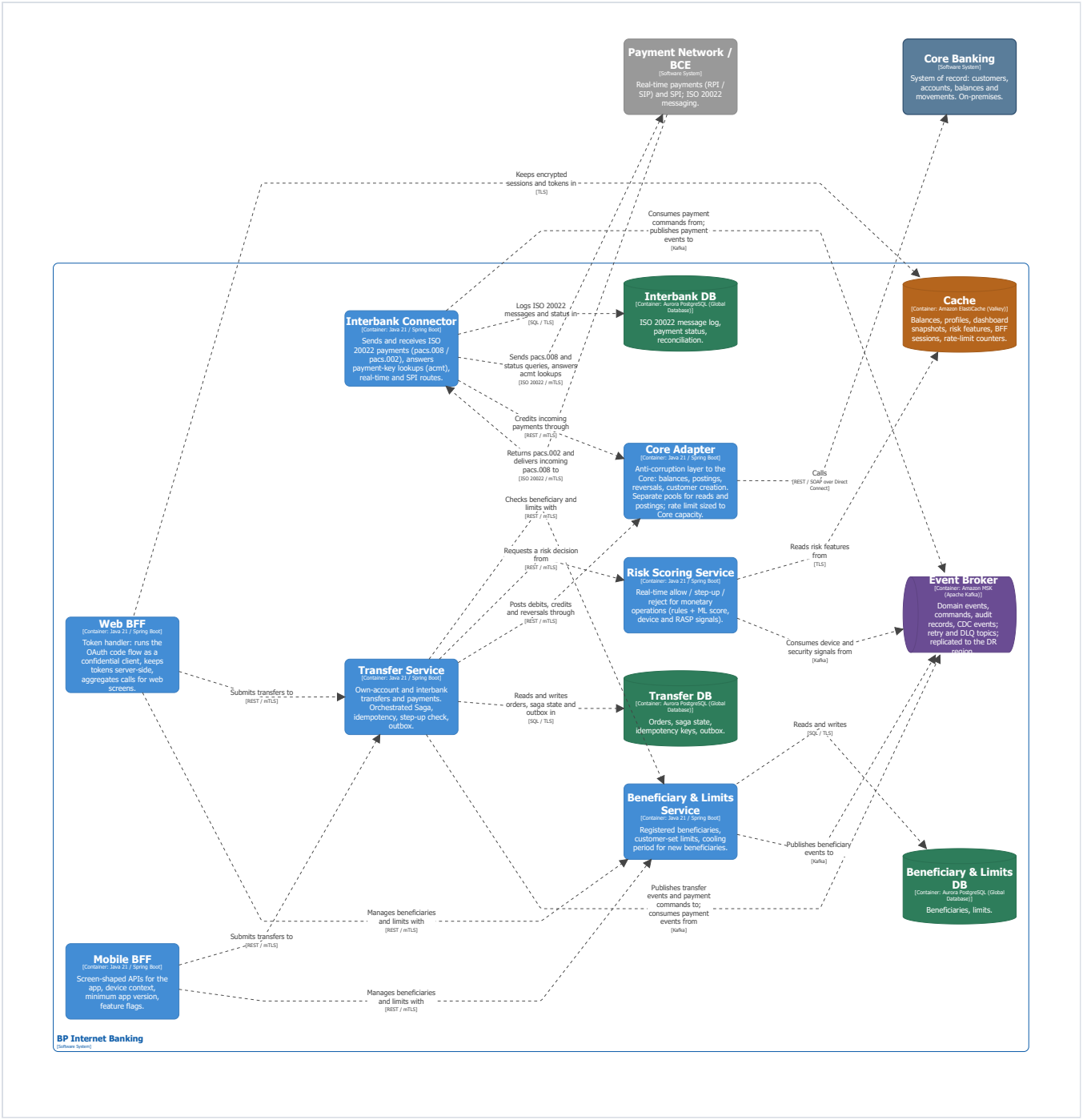


Figure 4.3c – Data access and frequent-client persistence

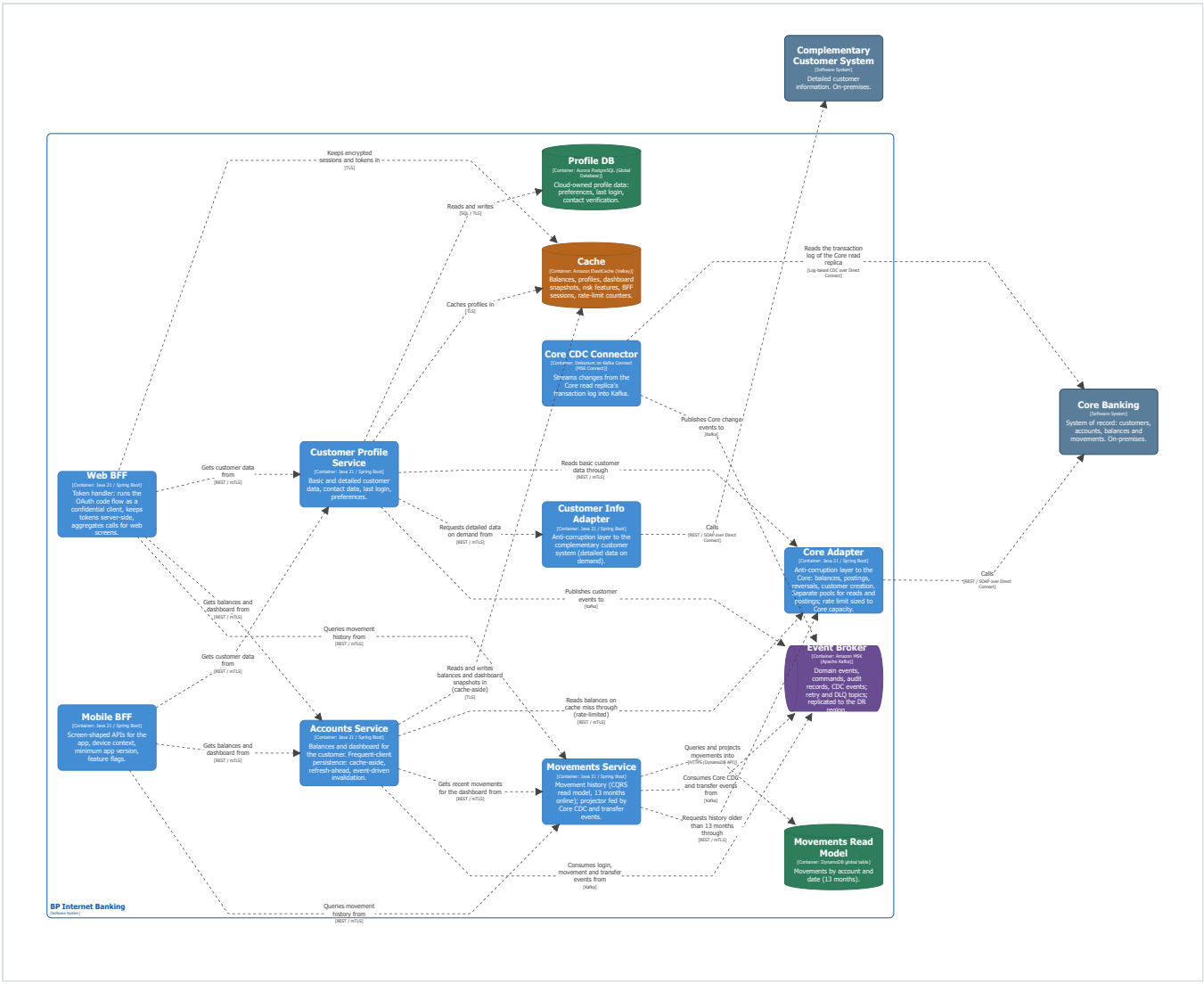


Figure 4.3d – Onboarding

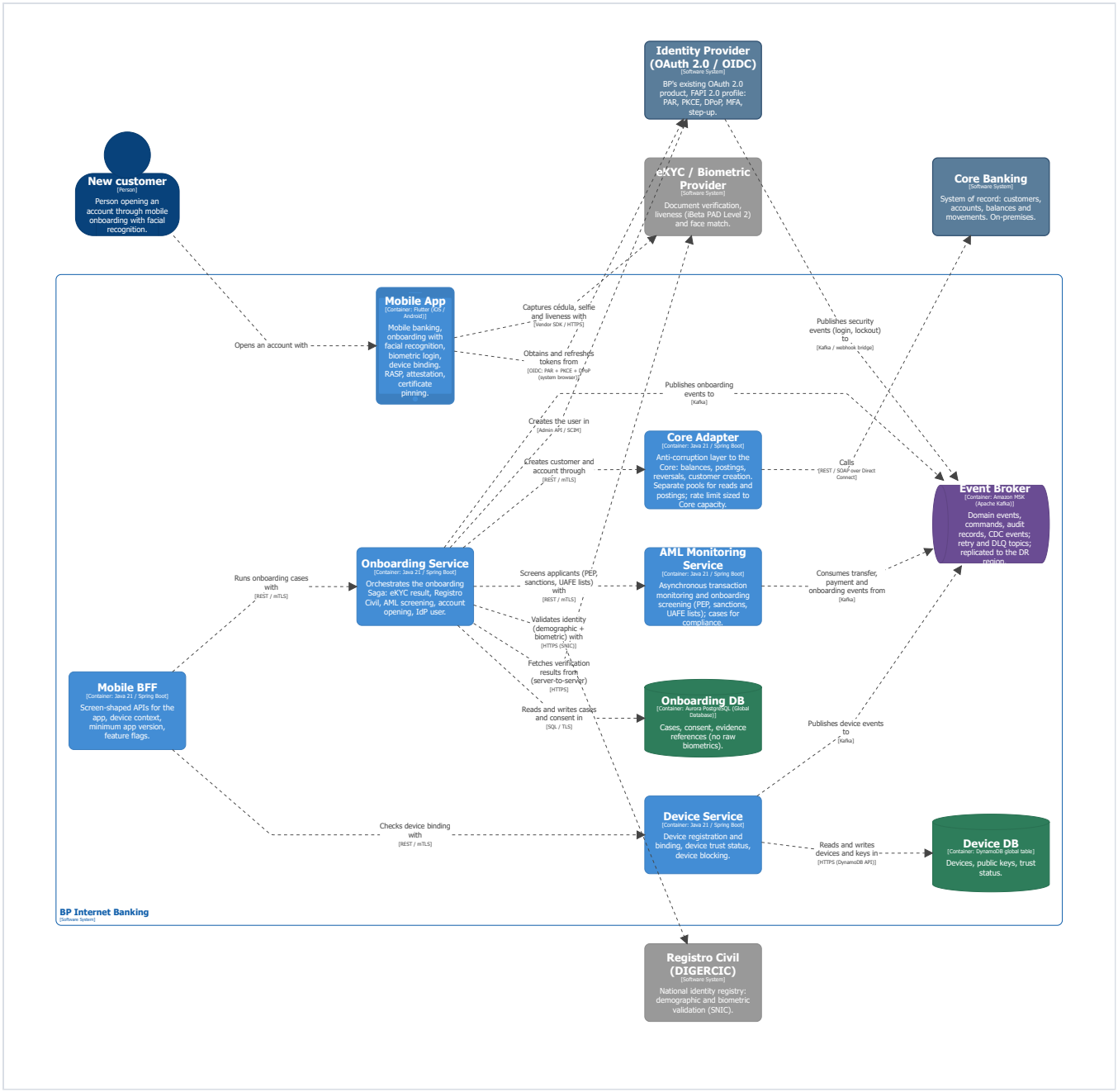
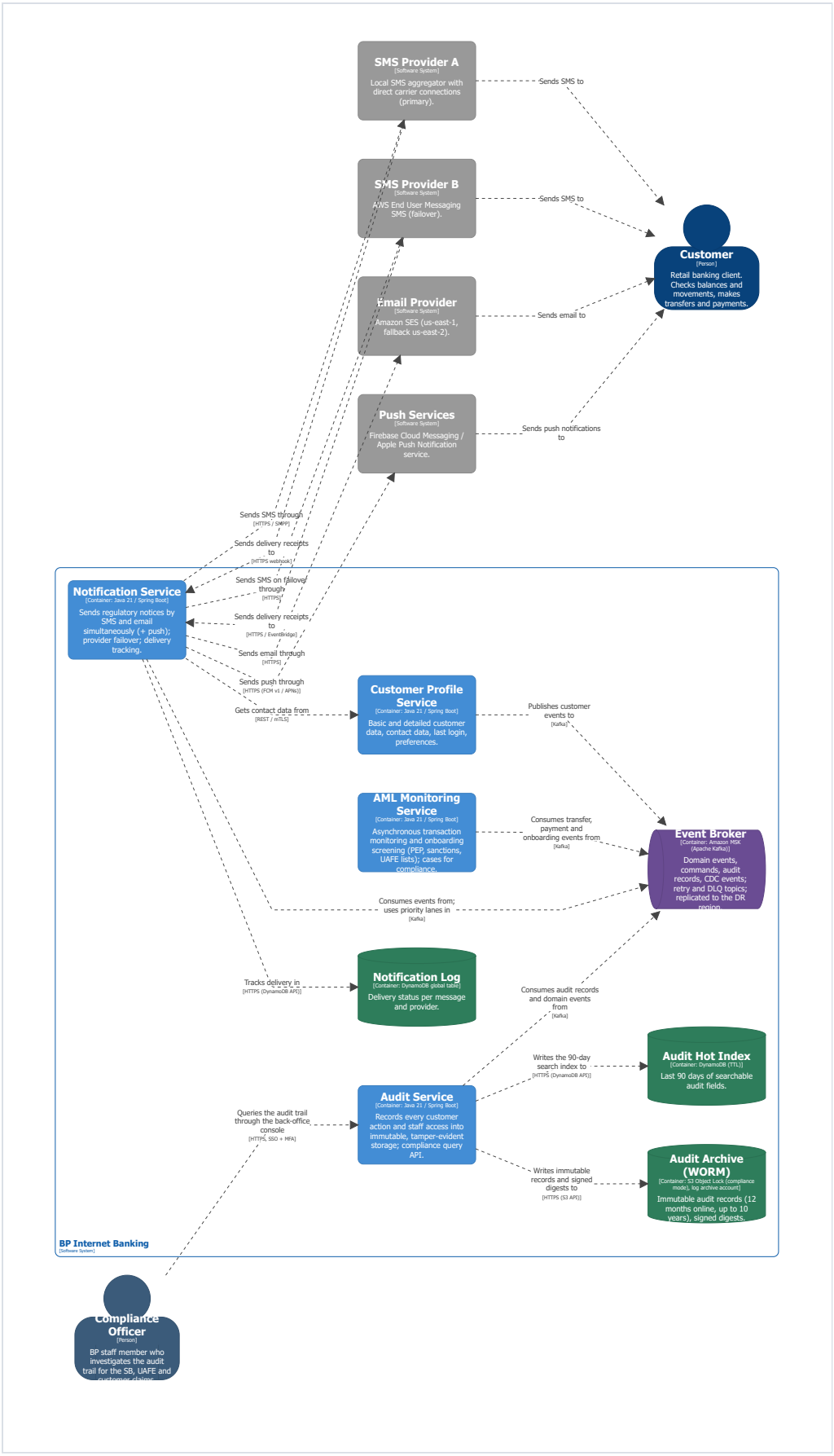


Figure 4.3e – Notifications and audit



Container	Technology	Responsibility
Web App (SPA)	Angular, S3 + CloudFront	Web UI; holds no tokens

Container	Technology	Responsibility
Mobile App	Flutter	Mobile UI, onboarding SDK, biometric login, device binding, RASP
Edge	CloudFront + WAF + Shield Advanced	TLS, WAF, bot control, DDoS, SPA hosting
API Gateway	Kong on EKS	Single entry point: JWT + DPoP + acr checks, rate limits, routing, access log
Web BFF / Mobile BFF	Java 21 / Spring Boot	Token handler (web); screen-shaped APIs, device context, app version (mobile)
14 domain services	Java 21 / Spring Boot on EKS	See section 4.1
Core CDC Connector	Debezium on MSK Connect	Streams Core changes into Kafka
Event Broker	Amazon MSK (Kafka)	Events, commands, audit records, CDC; replicated to DR
Cache	ElastiCache (Valkey)	Balances, profiles, dashboards, risk features, sessions
Transactional DBs	Aurora PostgreSQL Global Database	Transfer, Beneficiary, Onboarding, Profile, Interbank
Key-value / read models	DynamoDB global tables	Movements read model, Device, Notification log, Audit hot index
Audit Archive	S3 Object Lock (compliance mode)	Immutable audit records

4.4 Level 3: Components

Five containers are opened, chosen because they hold the patterns the statement asks about.

Figure 4.4a – API Gateway: plugin chain (token, DPoP, acr checks before routing)

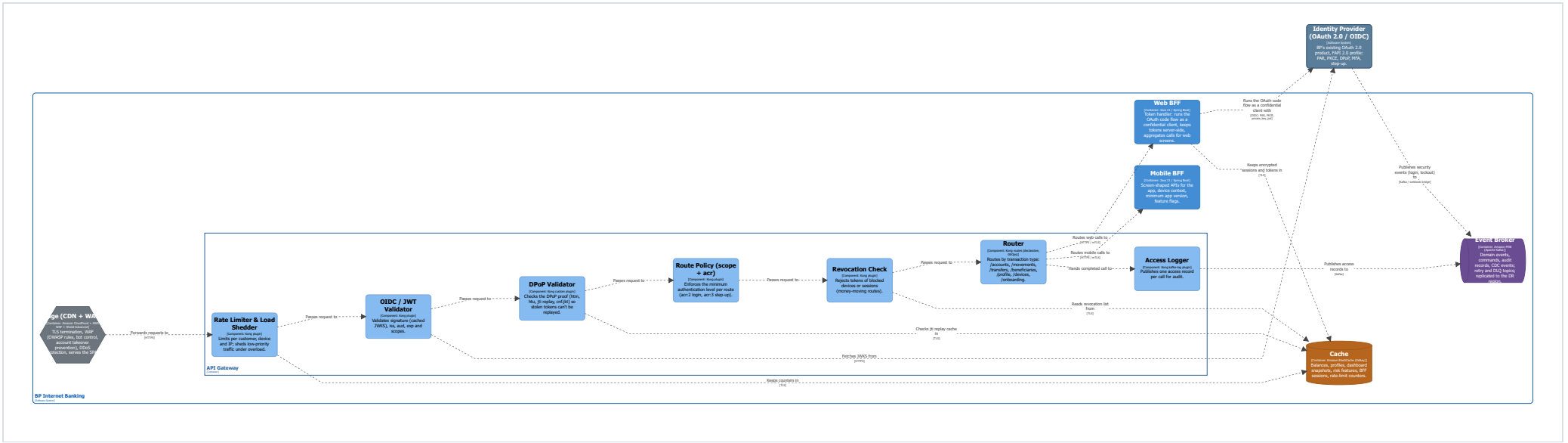


Figure 4.4b – Transfer Service: idempotency, step-up, Saga orchestrator, circuit-broken Core client, transactional outbox and inbox

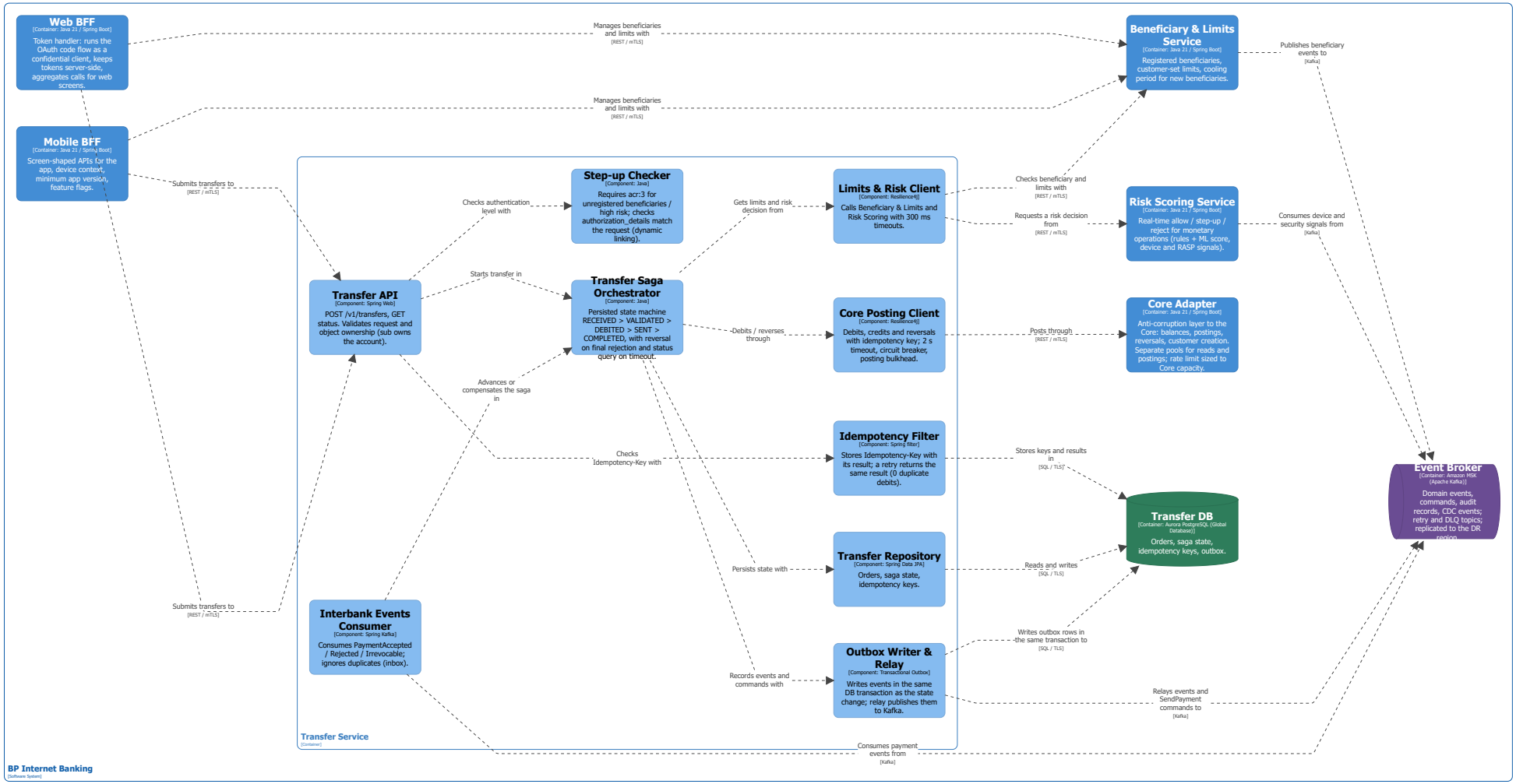


Figure 4.4c – Accounts Service: frequent-client persistence (cache-aside, refresh-ahead, event-driven invalidation)

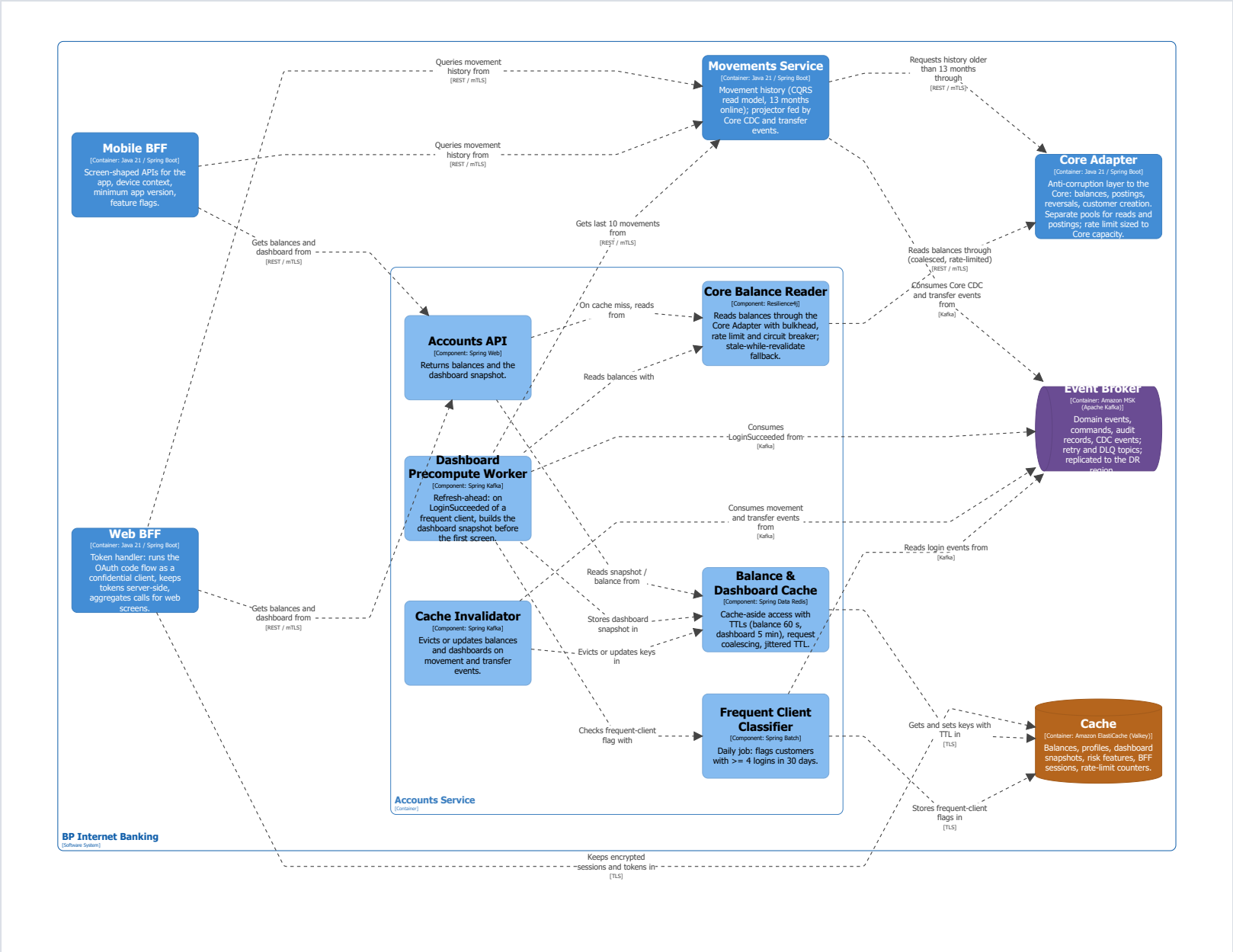


Figure 4.4d – Notification Service: policy, templates, priority lanes, SMS failover, delivery tracking

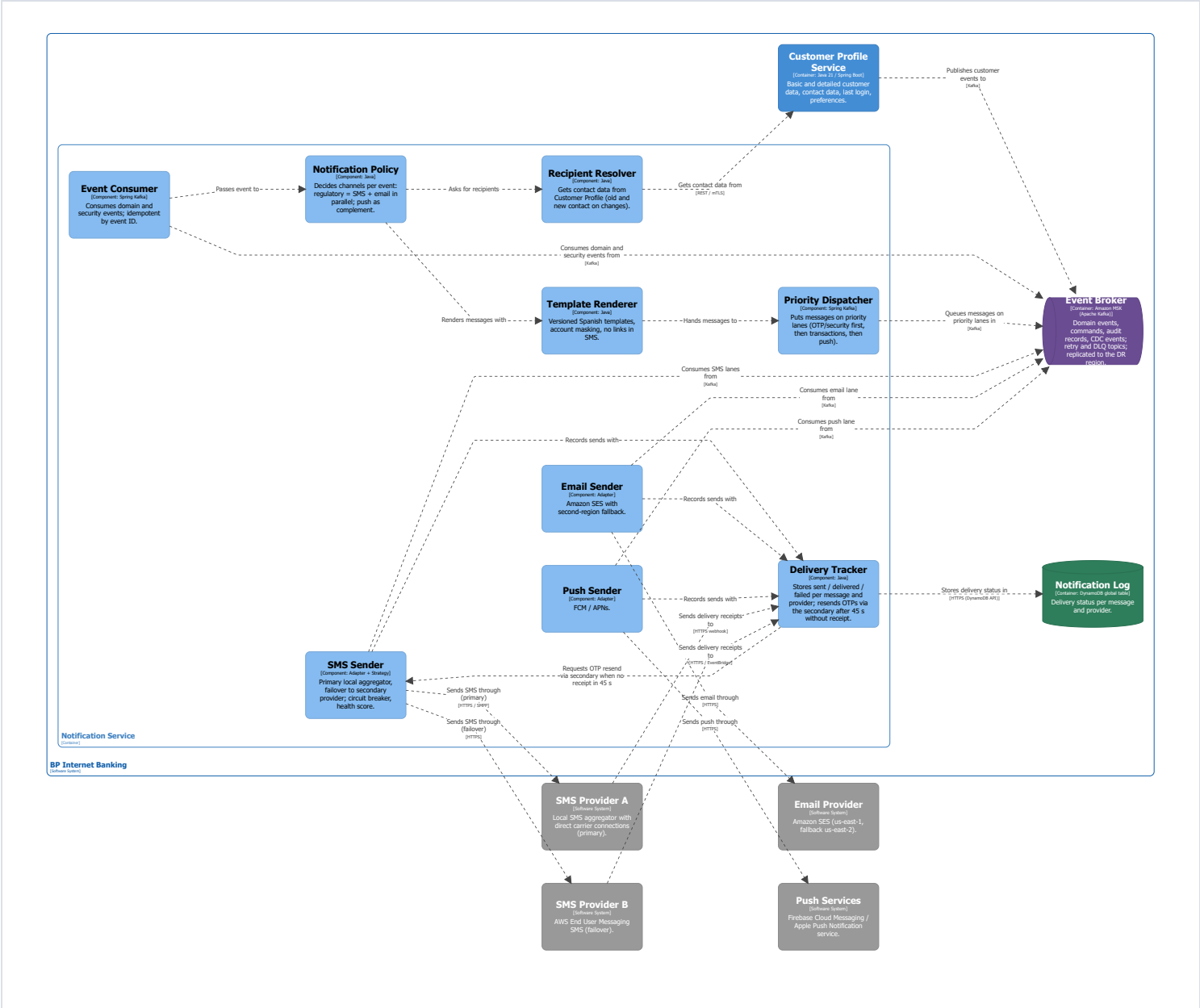
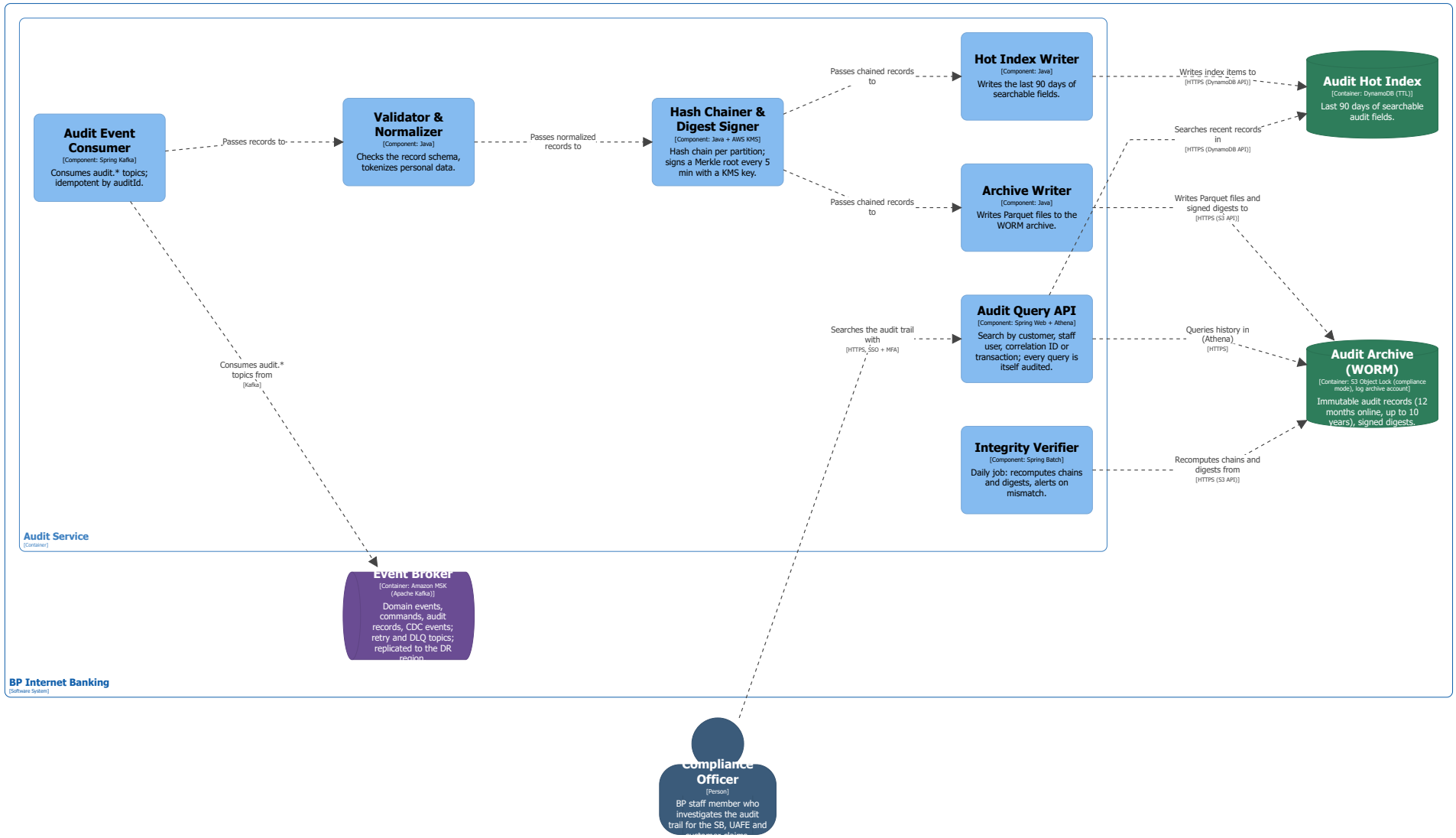
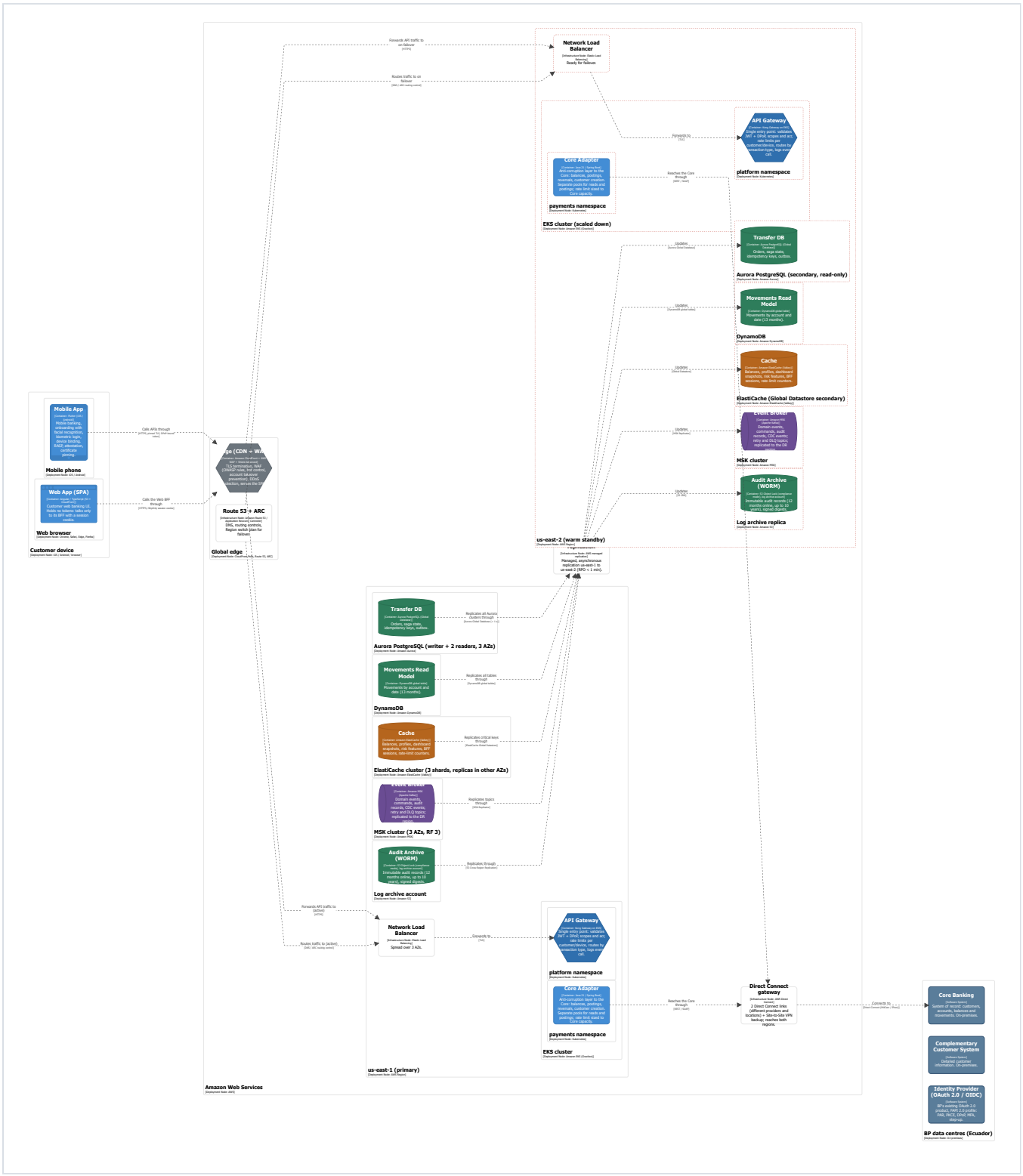


Figure 4.4e – Audit Service: idempotent consumer, normalization, hash chain, WORM archive and hot index



4.5 Infrastructure (deployment)

Figure 4.5 – Production on AWS: us-east-1 primary (3 AZs), us-east-2 warm standby, hybrid link to the Core



Region choice and latency (ADR-002, ADR-003). No AWS or Azure region exists in Ecuador, so the platform is hosted abroad under the SB cloud conditions. Round trips measured from Cuenca (8 Nov 2026): **us-east-1 ~90 ms**, us-east-2 ~103 ms, mx-central-1 ~118 ms, sa-east-1 ~157 ms (Ecuador's submarine cables route through Miami). With HTTP/2 connection reuse, edge TLS termination and cache hits, a frequent client's dashboard stays well under 500 ms. Calls to the on-premises Core cross ~90 ms, which is why reads are served from caches and read models.

One representative element per platform service is shown; node descriptions list everything deployed there. EKS namespaces: **platform** (Kong, CDC), **channels** (BFFs), **payments** (Transfer, Beneficiary & Limits, Risk, Interbank, Core Adapter), **customer** (Accounts, Movements, Profile, Device, Onboarding, Customer Info Adapter), **support** (Notification, Audit, AML).

4.6 Sequence diagrams

Figure 4.6a – Web login (Authorization Code + PKCE + PAR via the Web BFF)

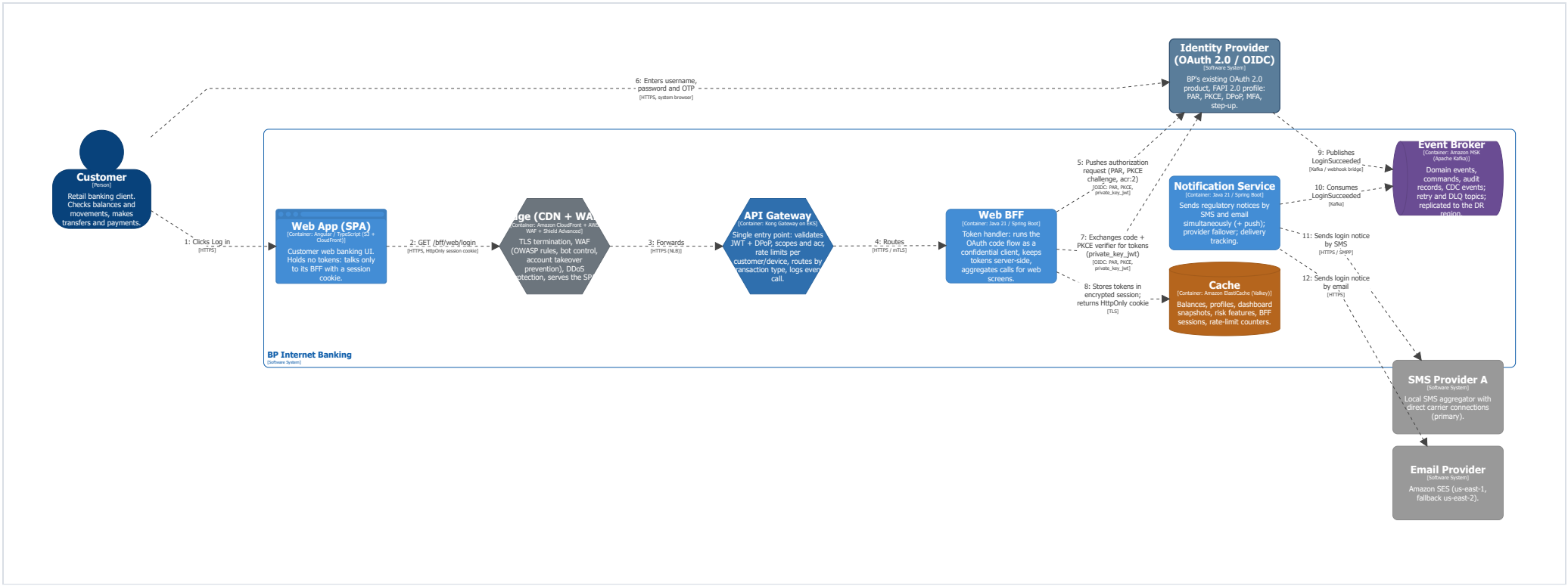


Figure 4.6b – Mobile biometric login with device-bound key and pre-built dashboard

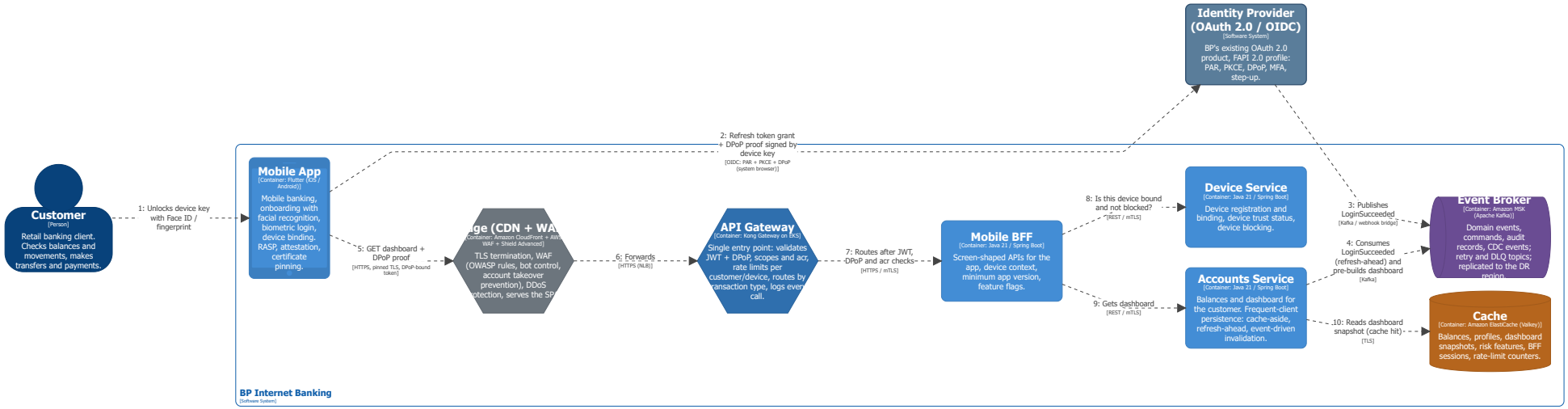


Figure 4.6c – Onboarding Saga with facial recognition

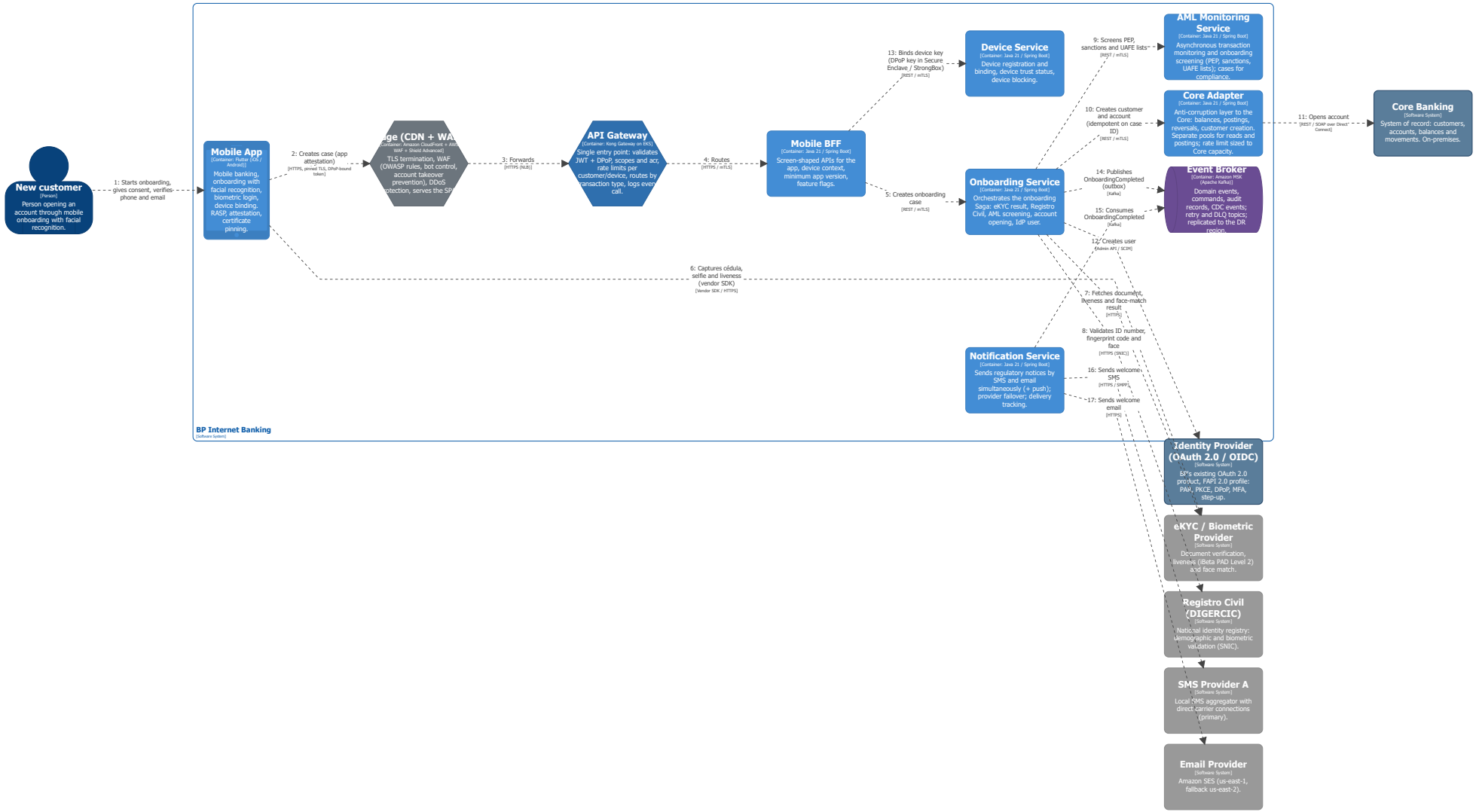
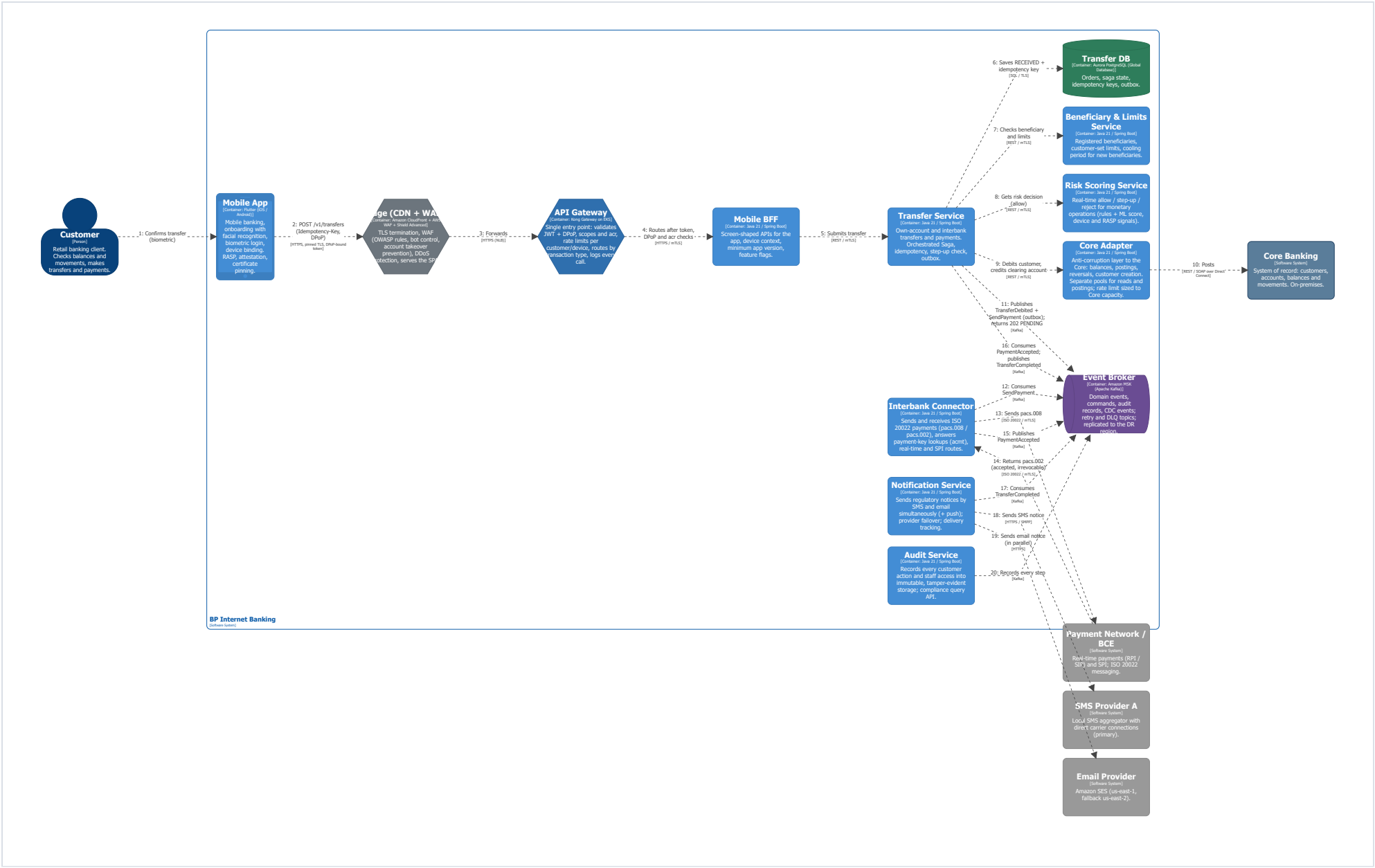


Figure 4.6d – Interbank real-time transfer (Saga, ISO 20022, notifications, audit)



5. Front-end and mobile architecture

ADR-005 – SPA: Angular (weighted score 4.65 vs React 4.00, Vue 3.95). (1) Opinionated, complete framework (router, reactive forms, HTTP interceptors, DI): every team builds the forms-heavy banking UI the same way. (2) Secure by default: values are sanitized before reaching the DOM; strict CSP and Trusted Types are supported, reducing XSS, the main threat to a banking SPA. (3) Predictable releases with 18 months of support per major version.

ADR-006 – Mobile: Flutter vs React Native → Flutter (4.20 vs 4.00).

Criterion (weight)	Flutter	React Native
Tamper resistance, RASP / attestation (20%)	5: AOT native code + obfuscation	4: Hermes bytecode
Biometrics and secure hardware (15%)	4	4
Performance and UI consistency on low-end Android (15%)	5: own rendering engine	4: New Architecture closed much of the gap
Banking adoption (15%)	4: Nubank	4: Coinbase
Talent in Ecuador (15%)	3	4
KYC SDK availability (10%)	4	4
Ecosystem and backing (10%)	4	4

Justifications: (1) stronger default protection of the binary, required by the SB anti-tampering rule and OWASP MASVS-RESILIENCE; (2) one rendering engine gives the same UI and accessibility behaviour on every device, fewer platform-specific defects; (3) proven at bank scale in Latin America (Nubank). Trade-off: smaller hiring pool.

Design rules

- **No tokens in the browser:** the Web BFF is a confidential OAuth client (token handler); the SPA receives only a `__Host-` HttpOnly, Secure, SameSite=Strict cookie (IETF *OAuth 2.0 for Browser-Based Apps*, BFF pattern). Strict CSP, Trusted Types, HSTS, anti-clickjacking headers.
- **One BFF per channel:** different security models (confidential web client vs native public client with DPoP) and release cycles.
- **Mobile security to OWASP MASVS L2 + R:** Keychain / Keystore with keys in **Secure Enclave / StrongBox**, **Play Integrity / App Attest** verified server-side, **commercial RASP** (root/jailbreak, hooking, emulator, overlay) feeding Risk Scoring, certificate **public-key pinning**, obfuscation, screenshot blocking on sensitive screens, forced update.
- **Micro-frontends rejected:** web is ~10% of traffic with few teams; one modular SPA (Nx, lazy-loaded features) instead, revisited above ~3 parallel web teams.
- Encrypted last-known dashboard on the device shown instantly "as of hh:mm"; idempotency key per transfer attempt; WCAG 2.2 AA; staged store rollouts and server-side feature flags.

6. Authentication and authorization

ADR-007 – Recommended flow: OAuth 2.0 Authorization Code + PKCE + PAR, FAPI 2.0 Security Profile, sender-constrained tokens. Implicit and Resource Owner Password grants are deprecated (RFC 9700) and disabled.

Channel	Client	Client authentication	Token binding	Token storage
Web	**Web BFF, confidential**	`private_key_jwt`	mTLS / DPoP from the BFF	Server-side session store only
Mobile	Public client, system browser (RFC 8252) + app attestation	PKCE + attestation	**DPoP** with a non-exportable key in secure hardware	Keychain / Keystore

Justifications: (1) **meets QAS-10** – a stolen token can't be replayed without the device key, lives 5 minutes, and can't move money without step-up; (2) **one security model for both channels**, validated identically at the gateway and reusable by new channels; (3) **FAPI is the international open-banking profile** (UK, Brazil, Australia), recognisable evidence for the SB; (4) **biometrics never leave the device**, so login doesn't process biometric data under the LOPDP.

Login methods after onboarding

Method	Factors	Use
Username + password + **OTP**	Knowledge + possession	Web and mobile; always available (non-biometric alternative)
Biometric login (Face ID / fingerprint)	Possession (device key) + inherence (biometric unlocks it)	Mobile; the key signs the DPoP proof of the refresh-token grant
Passkeys (FIDO2)	Possession + inherence	Optional for web; **not** used as the mobile factor because synced passkeys break "authorized devices only"

Step-up with dynamic linking: transfers to unregistered accounts, new beneficiaries, limit increases, contact changes and device registration require `acr:3`. The API answers `insufficient_user_authentication` (RFC 9470); the app re-authorizes with `authorization_details` (RFC 9396) containing amount and beneficiary; the resulting 2-minute, single-use token is valid only for that transaction. Web customers can approve in the mobile app (decoupled, CIBA).

Rules from the SB: lockout after 3 failures with immediate SMS + email; login notification; idle timeout (web 5 min); last-login date shown; old and new contacts notified on change; 24/7 device blocking.

Tokens: access 5 min (JWT, PS256/ES256 keys in KMS/HSM); refresh tokens rotated with reuse detection; ID token never sent to APIs.

Authorization: the **gateway** validates signature (cached JWKS), `iss/aud/exp`, the **DPoP proof** (htm, htu, jti replay, `cnf.jkt`), scopes and minimum `acr` per route, plus a revocation list on money-moving routes; **services** enforce object-level authorization (the account belongs to `sub`, OWASP API1) and limits; service-to-service traffic uses **mTLS** (Istio ambient).

The bank's identity product must support PAR, PKCE, `private_key_jwt`, DPoP, refresh rotation, `acr` / RAR, MFA, lockout events and JWKS; Keycloak, Ping/ForgeRock and Okta/Auth0 are candidate products with this feature set.

7. Onboarding with facial recognition

Checks: cédula document authenticity → **liveness** (iBeta PAD Level 2, ISO/IEC 30107-3) and injection-attack detection → **1:1 face match** with the document → **Registro Civil (DIGERCIC SNIC)** demographic + biometric validation (USD 0.30 per query) → age ≥ 18 → **PEP / sanctions / UAFE screening** → phone and email OTP → explicit **consent**.

ADR-008 – Buy a specialist eKYC platform instead of building on AWS Rekognition (which offers liveness but no cédula document verification or registry connector). Shortlist and PoC: **FacePhi** (preferred: documented Registro Civil Ecuador validation, Latin American banking track record) and **Jumio** (alternative: Ecuador registry check); iProov, Veridas, FaceTec, Onfido also evaluated. Justifications: (1) only specialist vendors cover document + registry + liveness together; (2) independently tested liveness and injection-attack defence give auditable evidence for QAS-11; (3) an anti-corruption layer keeps the vendor swappable. The bank holds the DIGERCIC contract itself as data controller and obligated KYC party.

Saga (orchestrated by the Onboarding Service)

Step	On failure
Consent, age, contact OTP	Stop; abandoned case deleted after 7 days
Document + liveness + face match (vendor SDK, results fetched server-to-server)	3 retries, then **assisted / branch onboarding**
Registro Civil validation	Registry down → pending + retry; mismatch → manual review
AML screening	Hit → **manual review** (never auto-reject)
Create customer and account in the Core (idempotent on case ID)	Retry; pending + alert
Create user in the IdP	Retry; account blocked until the user exists (no compensation)
Device binding + biometric login enrolment	Repeatable at next login
`OnboardingCompleted` via outbox	Audit, Notification, AML, Profile

Hand-off to authentication: the customer sets a password, the app generates the DPoP key in secure hardware and registers it with the Device Service (`acr:3` operation, notified by SMS + email), and biometric login is enrolled.

Biometric data: raw selfies and document images are deleted after the decision; **no face template is stored** (login uses on-device biometrics; recovery repeats liveness + registry); evidence (vendor transaction ID, scores, decision, reviewer) and the consent record go to the WORM audit store. DPIA before go-live and every 12 months; rejections reviewed by a person; no minors.

8. Integration layer and communication patterns

Interbank payments (Res. BCE-GG-007-2026): ISO 20022 `pacs.008` payment order, `pacs.002` accept / reject / irrevocability, `acmt` payment-key lookups; the ordering bank **debits first**, then sends; accepted payments are **irrevocable**; intra-network messaging due by **30 Nov 2026**. The Interbank Connector is two-way: it also **receives** incoming `pacs.008` (credit + `pacs.002` within the network timeout, a T1 path) and answers key lookups. Real-time route up to the cap, SPI route above it.

Decision	Choice	Main alternatives	Justifications
ADR-009 API Gateway	**Kong on EKS** (4.50)	Apigee 3.55, Amazon API Gateway 3.55, Azure APIM 2.70	DPoP / FAPI checks as plugins (AWS API Gateway would need a Lambda authorizer per call); flat cost at billions of requests; declarative, portable config in Git, identical in both regions
ADR-010 Messaging	**Amazon MSK (Kafka)** (4.70)	Confluent Cloud 4.55, SQS/SNS/EventBridge 3.15	Replay to rebuild read models and audit; per-account ordering; one backbone for events and CDC; data stays in the bank's account
ADR-011 Compute	**Amazon EKS** (4.80) + Lambda only for glue	ECS Fargate 3.65, Lambda 2.65	Portability and one platform for gateway, mesh and services; auto-healing and Karpenter/KEDA elasticity; steady low-latency load with long-lived Core and Kafka connections

Communication: REST/JSON (OpenAPI 3.1) for synchronous calls; Kafka for events and commands; gRPC deferred. **Service stack:** Java 21 + Spring Boot 3 (virtual threads, Resilience4j, OpenTelemetry). **Service mesh:** Istio ambient mode for mTLS and authorization policies; retries configured in one place only.

Patterns

Pattern	Where	Attribute
Timeout, retry with backoff + jitter (idempotent calls only)	All outbound calls	QAS-03
Circuit breaker	Core, complementary system, providers, eKYC, Registro Civil	QAS-03 / 04
Bulkhead	Core Adapter: separate pools for reads and postings	QAS-03 / 09
Rate limiting toward the Core	Token bucket sized to Core capacity	QAS-03
Idempotency key	Every money-moving request	QAS-05
Transactional outbox + **idempotent consumer (inbox)**	All event producers / consumers	Consistency, no lost audit events
Saga (orchestrated)	Transfers, onboarding	Consistency without 2PC
Anti-corruption layer	Core, complementary system, BCE network, eKYC	Decoupling
API composition / **BFF**	Screen aggregation	QAS-06
Strangler fig	Gateway routing during migration	Migration risk
Load shedding	Gateway sheds history exports before transfers	QAS-09

Interbank transfer Saga: `RECEIVED → VALIDATED → DEBITED → SENT → COMPLETED`; a final `pacs.002` rejection triggers a **reversal** (credit back); a timeout means **UNKNOWN** and a status query, never a blind reversal of a possibly irrevocable payment. The customer gets "accepted / pending" in < 3 s (QAS-08) and the final status asynchronously (FR-13).

9. Data access and frequent-client persistence

Principles: the Core is the system of record; cloud copies are read models; database per service; reads from the closest copy, writes to the owner; minimal personal data in caches and events.

ADR-012 – Data stores: Aurora PostgreSQL Global Database for transactional services (outbox + idempotency in one ACID transaction), **DynamoDB global tables** for high-volume key-value data and read models (movements ~4 B items, constant latency), **ElastiCache (Valkey)** for caching; only three data technologies. Justifications: right store per access pattern; managed cross-region replication meeting RPO 1 min; small technology set.

ADR-013 – Frequent-client persistence (statement §3.6). A frequent client has ≥ 4 logins in 30 days. Four patterns work together:

Pattern	Applied to	Effect
CQRS read model	Movements (DynamoDB, 13 months) fed by Core CDC + transfer events	History queries never touch the Core
Cache-aside	Balances (TTL 60 s), profiles (24 h), dashboard snapshot (5 min)	Most reads served from memory
Event-driven invalidation	Movement, transfer and incoming-payment events evict keys in seconds	No stale balances
Refresh-ahead + materialized view	On `LoginSucceeded` of a frequent client, a worker pre-builds the dashboard	First screen is a cache hit
Request coalescing, jittered TTL, stale-while-revalidate	All caches	No thundering herd; "as of hh:mm" fallback when the Core is down

Rejected: calling the Core every time (latency, Core load, no fault tolerance), cache-aside only (cold starts on payday), full Core replica (duplicates the system of record), device-only cache (not for web or new devices). Justifications: (1) meets QAS-06 and protects the Core even on payday; (2) each pattern covers another's weakness; (3) fault tolerance as a side effect, exactly the failure mode of the 2025 outages. A cached balance is **displayed only**; the Core checks the available balance when posting.

ADR-014 – Core CDC: log-based CDC with **Debezium** on a **Core read replica** (seconds of freshness, no Core code change, negligible load); nightly reconciliation; fallback: incremental polling for active customers only.

Data protection: KMS customer-managed keys per domain (crypto-shredding), TLS everywhere, application-level encryption of cédula, phone, email; tokenized identifiers in events and logs; masking; least-privilege IAM per service; synthetic or masked data outside production.

10. Audit solution

What is audited: every customer action including reads, security events, onboarding evidence and consent, **staff access to customer data**, and (separately) platform admin actions via CloudTrail. The record schema covers the SB minimum: date, time, amount, origin and destination accounts and banks, transaction number, notified phone and email, OTP channel, IP, device, correlation ID.

ADR-015 – Audit architecture. Capture: domain events via **transactional outbox**, gateway access records (Kong kafka-log), an `@Audited` **interceptor** for reads and back-office access, IdP security events → Kafka `audit.*` → **Audit Service** (validate, tokenize, **hash-chain**) → **S3 Object Lock in compliance mode** in the separate log-archive account (Parquet, queried with Athena) + **DynamoDB hot index** (90 days) → compliance console (every query itself audited, case number mandatory).

Storage option	Verdict
S3 Object Lock (compliance)	Chosen: nobody, not even root, can delete before retention ends; assessed for SEC 17a-4
DynamoDB	Hot index only (mutable)
OpenSearch	Not needed; costly at this volume, mutable
Aurora + pgAudit	Rejected: a DBA can change rows
Amazon QLDB	**Retired (end of support 31 Jul 2025)**

Justifications: (1) real immutability against insiders (QAS-12); (2) no event lost (outbox) and the critical path is never blocked (asynchronous, T2 service catches up from Kafka); (3) 10 years affordable (Parquet + Glacier Deep Archive); (4) **tamper evidence** on top: hash chain per partition, Merkle-root digest signed with KMS every 5 minutes, daily verification job, write-only cross-account access, SCPs that forbid disabling the lock. Full Event Sourcing was rejected as too complex for every service; outbox + pub/sub + append-only log gives the audit trail.

Retention: 12 months online (SB), 10 years archive for transaction and KYC/AML records, legal hold for claims. ~86 M records/day ≈ 3.2 TB/year compressed.

11. Notifications

ADR-016: the two regulatory notification systems are **SMS and email, sent in parallel** for every monetary transaction and login; push and an in-app inbox complement them.

Channel	Primary	Failover
SMS	**Local aggregator** with direct Claro / Movistar / CNT connections	**AWS End User Messaging SMS**
Email	**Amazon SES** us-east-1	SES us-east-2
Push	FCM / APNs	—

Justifications: (1) two independent systems for every regulatory notice; (2) SMS carries OTPs, so it has two providers with automatic failover (QAS-04, SB alternate-provider rule); (3) local aggregators give better price and delivery in Ecuador; (4) no retired services: **Amazon Pinpoint reached end of support on 30 Oct 2026**, only its continuing messaging APIs are used.

Design: event consumer → policy → recipient resolver → template renderer (masking, no links in SMS) → **priority lanes** (OTP/security before transactions) → senders with **adapter + strategy**, circuit breakers → delivery tracker with receipts; an OTP without delivery receipt in 45 s is resent through the secondary. SPF, DKIM, DMARC `p=reject` on the sending domain. Regulatory notices can't be switched off by the customer.

12. Security architecture

Principles: **Zero Trust** (NIST SP 800-207), defence in depth, least privilege, secure by default, assume breach.

Layer	Controls
Client	MASVS L2 + R, RASP, attestation, pinning, secure-hardware keys; BFF token handler, CSP
Edge	CloudFront only entry; **AWS WAF** managed rules, Bot Control, **Account Takeover Prevention**; **Shield Advanced**
API	Kong: JWT + DPoP, scopes, acr, rate limits per customer/device, schema validation
Identity	FAPI 2.0, 2-of-3 factors, step-up with dynamic linking, device binding, lockout; staff SSO with FIDO2 MFA
Service	Istio mTLS + authorization policies, object-level authorization, Risk Scoring before money moves
Network	Private subnets, VPC endpoints, **Network Firewall egress allowlist**, encrypted Direct Connect, deny-by-default NetworkPolicies
Data	KMS per domain, **CloudHSM** for HSM-bound keys (OTP seeds, network signing), field-level encryption, tokenization, Macie
Detection	CloudTrail, GuardDuty (EKS runtime), Security Hub, Inspector, **Security Lake → bank SIEM / 24×7 SOC**
Supply chain	SAST, SCA + SBOM, IaC policy checks, signed images (cosign) + admission control (Kyverno), DAST, annual pentest

Fraud: Risk Scoring returns allow / step-up / reject in ≤ 300 ms using device, RASP, behaviour, velocity and beneficiary signals (rules + ML score); cooling-period limits for new beneficiaries; beneficiary name shown from the `acmt` lookup; AML Monitoring watches mule patterns. The OWASP API Security Top 10 (2023) is mapped control by control in the step notes.

ADR-017 – Security tooling: AWS-native security services forwarding to the bank's existing SIEM through Security Lake (OCSF). Justifications: native coverage without agents; one SOC for cloud, Core and channels (supports SB incident deadlines); continuous compliance evidence for annual audits.

13. High availability, fault tolerance, DR and auto-healing

Tier	Services	Availability	RTO / RPO
T1	Edge, gateway, BFFs, Device, Accounts, Movements, Transfer, Beneficiary & Limits, Risk, Core Adapter, Interbank, Profile	99.99%	15 min / 1 min
T2	Onboarding, Notification, Audit, Customer Info Adapter	99.95%	1 h (Notification 15 min) / ~0 (events kept)
T3	AML, reporting, batch	99.9%	4 h / 1 h

Multi-AZ HA: ≥ 3 replicas per T1 service with topology spread and N+1 capacity; Aurora failover < 30 s; Kafka RF 3 / `min.insync.replicas=2`; ElastiCache Multi-AZ; **ARC zonal autoshift** away from impaired AZs.

ADR-018 – DR: warm standby (T1) in us-east-2, pilot light (rest of T2), backup & restore (T3). Failover is **decided by a person and executed by ARC Region switch plans** (Aurora Global Database failover and scaling, EKS scaling, routing): ≈ 12–15 min. Justifications: (1) meets RTO 15 min / RPO 1 min at ~12–15% extra cost, versus roughly double for active/active; (2) transfers need a single writer and the Core is a single site, so active/active adds conflict handling on money for little gain; (3) a different region satisfies the SB's remote-site rule and is drilled twice a year; (4) human-started, machine-executed failover avoids flapping. In-flight sagas resume from Aurora state, reconciled with the Core; failback is a planned zero-loss switchover.

Auto-healing: liveness/readiness probes (liveness never checks dependencies), Karpenter node replacement, HPA/KEDA scaling, **canary releases with automatic rollback**, GitOps drift correction, circuit breakers that close on recovery.

Graceful degradation: Core down → balances and movements "as of hh:mm", transfers rejected clearly; payment network down → interbank rejected before debit; one SMS provider down → failover; Notification or Audit down → events wait in Kafka; Risk Scoring down → fail closed for risky transfers.

Backups isolated from replication (ransomware): PITR 35 days, AWS Backup with **Vault Lock** in a separate account; quarterly restore tests. **Testing:** AWS FIS chaos experiments monthly (staging) and quarterly game days, AZ failure simulation, **full regional failover twice a year**, 3× load tests before payday seasons.

14. Monitoring and observability

ADR-019 – OpenTelemetry everywhere (ADOT collectors) → Amazon Managed Prometheus (metrics), X-Ray / Application Signals via OTLP (traces), CloudWatch Logs (logs), **Amazon Managed Grafana** (dashboards and alerts); security events → Security Lake → SIEM. Scores: AWS-managed open-source stack 4.25, self-hosted Grafana 4.10, Datadog 3.60, Dynatrace 3.50. Justifications: vendor-neutral instrumentation (and AWS's path now that the **X-Ray SDKs reach end of support on 25 Feb 2027**); telemetry stays in the bank's account (LOPDP); cost follows usage.

- **Correlation ID = trace ID**, propagated through HTTP **and Kafka headers**, logged, stored in audit records and returned to the app for support (QAS-13).
- **SLOs** per journey (login 99.95%, dashboard 99.99% available / 99.9% < 500 ms, transfers, **incoming payments answered within the network timeout 99.99%**, notifications < 60 s) with **error budgets** that pause releases when exhausted.
- **Burn-rate alerting** (multi-window) every minute → PagerDuty; Opsgenie (end of support Apr 2027) and AWS Incident Manager (closed to new customers) avoided.
- **Business dashboards:** transfers per minute vs last week, pending/unknown interbank age, onboarding funnel, notification delivery and cost per provider, Core latency and CDC lag, cache hit ratio, DR replication lag.
- Synthetic journeys every minute from outside AWS (ideally from Ecuador); real-user monitoring and crash reporting; no personal data in telemetry (redaction in the collector), tail sampling.

15. Operational excellence

- **ADR-020 – Terraform** (or the open-source fork **OpenTofu**): one tool for AWS and non-AWS resources (Kong, Grafana, PagerDuty, GitHub), plan-before-apply as change evidence, portable skills.
- **Environments:** dev (ephemeral PR environments), QA (vendor sandboxes), staging (production-like: performance, chaos, DR rehearsal, pentest), production in two regions; same image digest promoted; real data never leaves production.
- **Pipeline:** build → tests → SAST, secret scan → SCA + SBOM → contract tests (OpenAPI, Kafka schemas) → image scan and signing → GitOps (Argo CD) → **canary 5% → 25% → 50% → 100% with automatic rollback** on SLO breach; mobile via Fastlane with staged store rollouts; feature flags (OpenFeature / AppConfig) and kill switches; expand/contract database migrations; release freezes around payday.
- **Change management:** bank CAB kept, with **pre-approved standard changes** when the pipeline proves its controls; evidence stored automatically for auditors and the SB.
- Runbooks per alert, incident process with SB/SPDP deadlines, blameless post-incident reviews, operational readiness reviews, DORA metrics, stream-aligned teams plus platform, security and SRE teams, and a **golden-path service template**.

16. Cost management

Order-of-magnitude monthly estimate (AWS list prices, us-east-1; to be confirmed in the AWS Pricing Calculator):

Area	USD / month
Compute (EKS, Graviton, Savings Plans on baseline, warm standby)	~8,000
Aurora PostgreSQL (5 clusters + Global Database)	~6,500
DynamoDB (read model, device, notification, audit index)	~5,000
ElastiCache + Global Datastore	~2,500
MSK (primary + DR + Replicator)	~4,500
Networking (Direct Connect, Transit Gateway, NAT, Network Firewall, endpoints)	~7,000
Edge (CloudFront, WAF, Shield Advanced, Bot Control / ATP on login)	~15,000–22,000
Security (GuardDuty, Security Hub, Inspector, Macie, Security Lake, KMS, CloudHSM)	~10,000
Observability	~10,000
Storage and backup	~1,500
Email (SES, ~164 M)	~16,000
Production	**≈ 85,000–95,000**
Non-production (scheduled shutdown)	≈ 20,000–25,000
AWS total	**≈ 105,000–120,000 (≈ USD 1.3–1.4 M / year)**

Outside AWS: SMS ≈ **USD 0.8–3.3 M / month** depending on the negotiated price (7–30× the whole AWS bill), Registro Civil ~USD 15 k, eKYC vendor and licences by quote. **DR cost:** warm standby ≈ USD 10–12 k / month vs ≈ USD 50–60 k for active/active.

ADR-021 – Cost optimization: commit to the steady baseline (Savings Plans, reserved instances), scale everything above it automatically, run on **Graviton**, and manage cost with **FinOps** (mandatory tags checked in the pipeline, budgets and anomaly detection, cost per user / transaction / SMS, monthly reviews). Justifications: a clear baseline with predictable peaks; better price-performance on ARM; cost made visible to each team. Main levers: SMS price negotiation and push where the SB allows it, non-production shutdown, Spot for CI and batch, WAF bot/ATP scoped to login, log classes and tail sampling, S3 lifecycle for audit.

17. Architecture decision log

Every ADR records its context, the alternatives evaluated, the decision and at least two justifications; the full text is in the repository step notes (roadmap-steps/).

ADR	Decision	Alternatives evaluated	Justifications (summary)
001	Structurizr DSL for C4	draw.io, C4-PlantUML, IcePanel	One model → consistent views; diagrams as code in Git; context, container, component, deployment and dynamic views in one tool
002	AWS	Azure	Lowest measured latency (~90 ms us-east-1); managed multi-region data services for RPO/RTO; Financial Services Lens
003	us-east-1 primary, us-east-2 DR	São Paulo; us-east-1 + sa-east-1; single region	Latency; DR region with separate risks (SB) and acceptable latency (~103 ms)
004	Microservices by bounded context (coarse-grained)	Modular monolith; fine-grained microservices	Independent scaling of reads and payday transfers; fault isolation; data boundaries per compliance rules; reuse
005	Angular SPA	React, Vue	Complete opinionated framework; secure by default; predictable support
006	Flutter	React Native	Tamper resistance; consistent UI on low-end devices; proven in LATAM banking
007	Code + PKCE + PAR, FAPI 2.0, DPoP, BFF token handler	ROPC, Implicit, tokens in browser storage, synced passkeys on mobile	Stolen-token resistance; one model for both channels; open-banking standard; biometrics stay on device
008	Specialist eKYC vendor (FacePhi / Jumio PoC) + Registro Civil	Build on Rekognition; other vendors	Document + registry + liveness together; certified liveness evidence; swappable via ACL
009	Kong Gateway on EKS	Amazon API Gateway, Apigee, Azure APIM	DPoP/FAPI plugins; flat cost at volume; portable declarative config
010	Amazon MSK (Kafka)	Confluent Cloud, SQS/SNS/EventBridge	Replay; per-key ordering; CDC on the same backbone
011	Amazon EKS	ECS Fargate, Lambda	Portability and one platform; auto-healing and elasticity; steady low-latency load
012	Aurora PostgreSQL + DynamoDB + ElastiCache	Single shared DB, DocumentDB, OpenSearch	Right store per access pattern; managed cross-region replication; few technologies
013	CQRS + cache-aside + event invalidation + refresh-ahead	Direct Core calls, cache-aside only, full replica, device-only cache	QAS-06 and Core protection; patterns cover each other; fault tolerance
014	Debezium CDC on Core read replica	Vendor replication, polling, batch files	Fresh data without Core changes; low load; same event platform
015	Outbox + Kafka → S3 Object Lock + hot index, hash chain	DynamoDB, OpenSearch, Aurora + pgAudit, QLDB (retired), Event Sourcing	Real immutability; no lost events, non-blocking; affordable 10 years; tamper evidence
016	SMS + email in parallel, 2 SMS providers, SES	Twilio, Infobip, SendGrid, Azure ACS, Pinpoint (retired)	Two independent systems; OTP failover; local price/delivery; no retired services
017	AWS-native security services → bank SIEM	Third-party CNAPP / SIEM only	Native coverage; one SOC; continuous compliance evidence
018	Warm standby DR, ARC Region switch	Backup & restore, pilot light, active/active	RTO/RPO at reasonable cost; single writer for money; SB remote site; no flapping

ADR	Decision	Alternatives evaluated	Justifications (summary)
019	OpenTelemetry + AWS-managed Prometheus/Grafana	Datadog, Dynatrace, self-hosted Grafana	Vendor neutrality; telemetry in account; cost follows usage
020	Terraform / OpenTofu	AWS CDK, CloudFormation	Multi-provider; plan as change evidence; portability
021	Baseline commitments + autoscaling + Graviton + FinOps	On-demand only; full commitment	Predictable baseline; price-performance; visible cost per team

18. Risks, open questions and future evolution

Risk / open question	Mitigation
SMS cost at regulatory volume	Negotiate local price; ask the SB whether push to a bound device counts as "mobile messaging"
Core capacity and availability (outside our control)	Caching, CQRS, rate limiting, bulkheads, graceful degradation; Core modernization roadmap
Identity product lacks FAPI features (PAR, DPoP, RAR)	Add them at the gateway or choose a FAPI-certified product
Identity product and payment-network link must be reachable from both regions	Stated as a DR prerequisite
eKYC vendor accuracy on real cédulas and low-end phones	PoC with FacePhi and Jumio before contract
Registro Civil biometric data leak investigation (2026)	Registry match always combined with live liveness
Regulation details to confirm (SB article numbers, 10-year AML retention)	Legal review before go-live
Hyperscaler region in Chile/Colombia may open	Re-evaluate region choice (AWS Chile region announced)
Team skills (Flutter, Kubernetes, Kafka)	Platform team, golden paths, training

Future evolution: new channels (chatbot, DEUNA, open finance APIs) reuse the gateway, identity, notifications and audit (QAS-15); bill payments and loans added as new bounded contexts; gradual migration of Core functions behind the anti-corruption layer (strangler fig); active/active for read journeys if latency or availability targets rise.

19. Requirement and grading-criteria traceability

Grading criterion	Where answered
1. Requirements met and justified	§2, §17 (21 ADRs, ≥ 2 justifications each)
2. Quality and depth of C4 diagrams	§4 (context, 5 container views, 5 component views, deployment, 4 sequences)
3. Separation of responsibilities and decoupling	§4.1, §8 (bounded contexts, BFFs, ACLs, events)
4. Architecture patterns	§8, §9, §10, §11 (Saga, outbox, inbox, circuit breaker, bulkhead, CQRS, cache-aside, refresh-ahead, ACL, BFF, strangler fig)
5. Integration with external services	§7, §8, §11 (Core, complementary system, BCE ISO 20022, eKYC, Registro Civil, SMS/email/push)
6. Front-end and mobile architecture	§5 (Angular, Flutter vs React Native, MASVS)
7. Data access architecture	§9
8. Cloud knowledge (AWS)	§4.5, §8, §9, §13
9. Cost management	§16
10. Authentication architecture	§6
11. Onboarding integration	§7, Figure 4.6c
12. Audit solution	§10, Figure 4.4e
13. Banking regulation and security standards	§3, §12
14. High availability and fault tolerance	§13
15. Monitoring	§14

Statement checklist item	Section
C4 context, containers, components; infrastructure diagram	§4.2–4.5
2 mobile frameworks, choice justified	§5
Recommended OAuth 2.0 flow with justification	§6
Onboarding with facial recognition in authN/authZ, tools suggested	§6, §7
Login methods after onboarding	§6
≥ 2 notification channels / providers	§11
Integration with the Core and the complementary system	§8, §9
API Gateway + base services + additional services	§4.1, §8
Audit and frequent-client persistence based on design patterns	§9, §10
Regulation	§3
HA, fault tolerance, DR, security, monitoring, operational excellence, auto-healing	§12–15
Low latency and cost management	§4.5, §9, §16
≥ 2 justifications per decision with alternatives	§17

Repository: C4 model (architecture/workspace.dsl), rendered diagrams (PNG and SVG), and the detailed step notes with sources (roadmap-steps/).